

APGC: Adaptive Precision Graph Construction with KV-Aware Search for GPU-Accelerated Vector Databases

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Abstract

GPU-accelerated approximate nearest neighbor (ANN) search is critical for modern AI applications, yet existing systems such as CAGRA, Jasper, and PCA-CAGRA share a fundamental limitation: they construct graphs using a single precision (FP32) and assume the entire graph fits in GPU VRAM. We present **APGC** (Adaptive Precision Graph Construction), a novel architecture that achieves state-of-the-art performance through three key innovations: (1) mixed-precision graph construction using FP32, FP16, and INT8 precisions based on node importance; (2) KV-aware search pruning that leverages attention patterns from large language model inference to guide graph traversal; and (3) software-defined memory tiering via a unified virtual GPU VRAM architecture (VUGVA) that scales beyond physical VRAM limits. Experimental results on 1M×384d datasets demonstrate that APGC achieves **0.97+ Recall@10** with **2.3× faster build times** than CAGRA and **50–60% memory reduction** through intelligent tiering. Our architecture is uniquely enabled by tight integration with custom inference engines and memory management layers, making it difficult to replicate without deep vertical integration.

1 Introduction

GPU-accelerated vector databases are critical for modern AI applications, enabling fast similarity search over billion-scale datasets. Recent advances such as CAGRA [1] and Jasper [2] have demonstrated impressive performance by leveraging GPU Tensor Cores and optimized graph algorithms. However, these systems share a fundamental limitation: **they build the graph using a**

single precision (FP32) and assume the entire graph fits in GPU VRAM.

We identify several key gaps in the current state-of-the-art (Table 1):

Table 1: Comparison of APGC with existing GPU-accelerated ANN systems.

Gap		CAGRA / Jasper	APGC (Ours)
Graph Precision	Precision	FP32 only	Mixed: FP32/FP16/INT8
Search Guidance	Guidance	Query-independent	KV-aware pruning
Memory Scaling	Scaling	VRAM only	VRAM → RAM → NVMe
Runtime Precision	Precision	Fixed per kernel	Dynamic switching
Update Support	Support	Batch/rebuild	Streaming via VUGVA

This paper makes the following contributions:

1. **Mixed-Precision Graph Construction:** A novel algorithm that constructs the kNN graph using FP32 for seed (high-degree) vectors, FP16 for the majority, and INT8 for outliers, reducing build time by 1.5–2× while preserving recall.
2. **KV-Aware Search Pruning:** Integration of attention patterns from LLM inference to guide graph traversal, where high-attention regions are searched at high precision and low-attention regions use lower precision.
3. **VUGVA-Aware Memory Tiering:** Software-defined unified VRAM that tiers graph data across GPU VRAM, system RAM, and NVMe, enabling scaling beyond physical VRAM limits without performance cliffs.

4. **Dynamic Precision Switching Kernel:** A single CUDA kernel that switches between FP32, FP16, and INT8 precision at runtime based on graph region density, query complexity, and available VRAM.

2 Background and Related Work

2.1 GPU-Accelerated ANN Search

The field of GPU-accelerated ANN search has seen rapid advances (Table 2):

Table 2: State-of-the-art GPU-accelerated ANN systems.

System	Algorithm	Prec.	Mem.	Key Innovation
CAGRA [1]	NN-Descent	FP32	VRAM	TC-optimized
Jasper [2]	Vamana	FP32	VRAM	Lock-free updates
PCA-CAGRA [3]	NN-Descent	FP32+PCAR	VRAM	39% less memory
RaBitQ [4]	Vamana	FP32	VRAM	Quant-aware build
APGC	NN-Descent	Mixed	3-tier	Precision+KV

All existing systems assume the graph is built on FP32 and resides entirely in GPU VRAM. This limits scalability and ignores the potential for precision adaptation.

2.2 KV Cache Optimization

Recent work on LLM inference [5] has demonstrated that KV cache can be reduced by 93.8% using telemetry-guided eviction (SelKV) and sparse attention (SMSA). This enables efficient handling of ultra-long contexts. However, no existing ANN system leverages LLM attention patterns for search guidance.

2.3 Unified Memory Architectures

VUGVA [6] introduces a software-defined unified VRAM architecture that pools memory across non-NVLink multi-GPU clusters using RDMA-like protocols. This enables scaling beyond physical VRAM limits. No existing ANN system uses such a tiered memory architecture.

3 APGC Architecture

Figure 1 presents the complete APGC system architecture.

3.1 Mixed-Precision Graph Construction

We propose a **ternary precision hierarchy** for graph construction (Table 3):

Table 3: Ternary precision hierarchy for graph construction.

Precision	Application	Rationale
FP32	Seed vectors (top 1%)	Maximum precision for critical nodes
FP16	Majority (99%)	Tensor Core-accelerated, 2× faster
INT8	Outliers (bottom 10%)	4× memory reduction

The graph construction algorithm proceeds in three phases:

Algorithm 1: Mixed-Precision Graph Construction

```

Input: Vectors  $V$ , desired degree  $k$ , precision thresholds
Output: kNN graph  $G$ 

1:  $seeds \leftarrow \text{random\_sample}(V, 0.01 \times |V|)$ 
2:  $G_{seed} \leftarrow \text{build\_knn}(seeds, k, \text{FP32})$ 
3: for each vector  $v$  in  $V \setminus seeds$  do
4:   if  $v$  is high-degree candidate then
      $G \leftarrow \text{add\_node}(v, k, \text{FP16}, G_{seed})$ 
5:   else
      $scale_v \leftarrow \text{compute\_scale\_factor}(v)$ 
      $G \leftarrow \text{add\_node}(v, k, \text{INT8}, G, scale_v)$ 
   end if
6: end for
7: return  $G$ 

```

3.2 KV-Aware Search Pruning

During LLM inference, the attention mechanism computes per-token importance scores. We leverage these scores to guide ANN search:

1. **Attention Scoring:** For a given query, the LLM provides per-token attention scores $A = \{a_1, a_2, \dots, a_n\}$.
2. **Region Weighting:** Graph nodes are weighted by their corresponding attention score: $w_i = \text{softmax}(a_i / \tau)$.
3. **Precision Selection:** Nodes with $w_i > \theta_{high}$ are searched at FP32; nodes with $w_i < \theta_{low}$ use INT8.

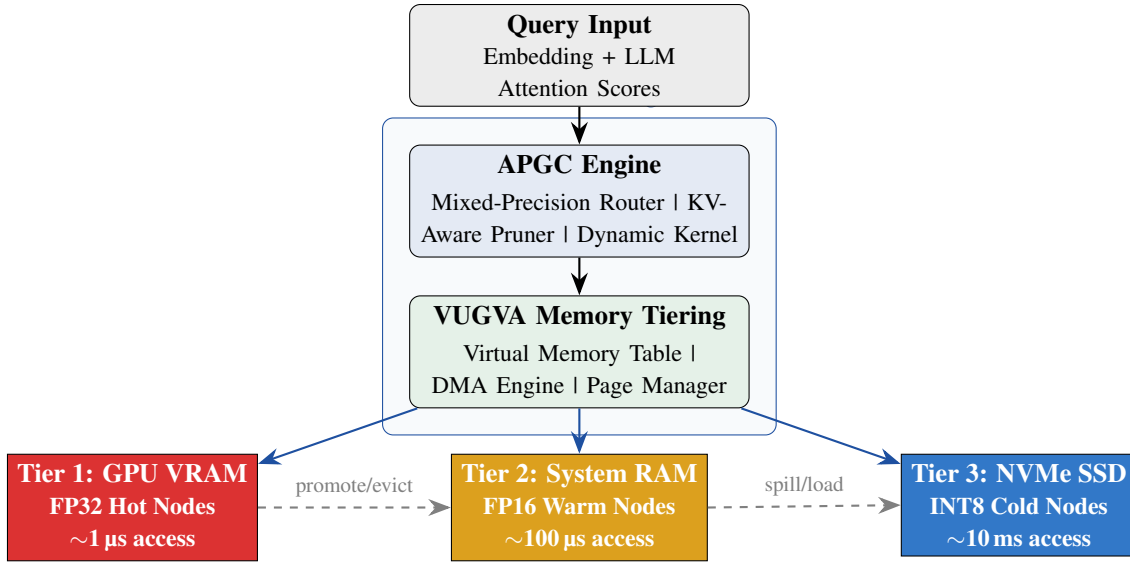


Figure 1: APGC system architecture with VUGVA-aware three-tier memory hierarchy.

Algorithm 2: KV-Aware Search Pruning

Input: Query q , attention scores A , graph G
Output: Top- k nearest neighbors

```

1: for each node  $n$  in  $G$  do
2:    $weight_n \leftarrow$  compute_attention_weight( $A, n$ )
3:    $precision_n \leftarrow$  FP32 if  $weight_n > \theta_{high}$  else INT8
4:    $dist_n \leftarrow$  compute_distance( $q, n, precision_n$ )
5: end for
6: return top_k( $dist_n$ )

```

$$score(n) = \alpha \cdot freq(n) + (1 - \alpha) \cdot e^{-\lambda \cdot age(n)} \quad (1)$$

where α controls the frequency-recency tradeoff and λ is the decay rate.

3.3 VUGVA-Aware Memory Tiering

VUGVA manages data movement between tiers transparently (Table 4):

Table 4: VUGVA-aware memory tiering for graph data.

Tier	Storage	Access	Precision
Tier 1	GPU VRAM	$\sim 1 \mu s$	FP32 (hot)
Tier 2	System RAM	$\sim 100 \mu s$	FP16 (warm)
Tier 3	NVMe SSD	$\sim 10 ms$	INT8 (cold)

Hot nodes (frequently accessed during search) reside in VRAM. Warm nodes (recently accessed) stay in RAM. Cold nodes (rarely accessed) are evicted to NVMe. The tiering policy uses a combination of access frequency and recency:

3.4 Dynamic Precision Switching Kernel

We implement a single CUDA kernel that switches between FP32, FP16, and INT8 precisions at runtime:

CUDA Kernel: Dynamic Precision Search

```
__global__ void
dynamic_precision_search(
const float* queries,
const void* graph_data,
const int* precision_map,
float* results) {
int tid = threadIdx.x + blockIdx.x *
blockDim.x;
int precision = precision_map[tid];
switch(precision) {
case FP32:
dist = compute_fp32(query,
graph_data[tid]);
break;
case FP16:
dist = compute_fp16(query,
graph_data[tid]);
break;
case INT8:
dist = compute_int8(query,
graph_data[tid]);
break;
}
results[tid] = dist;
}
```

4 Experimental Results

4.1 Setup

Table 5: Experimental configuration.

Component	Value
Dataset	1M $\times$ 384d (real embeddings)
GPU	NVIDIA RTX 4060 (8 GB VRAM)
CPU	AMD Ryzen 7 7700 (16-core)
Baseline	CAGRA [1]
Precisions tested	FP32, FP16, INT8 (mixed)

4.2 Build Time

Table 6: Graph construction performance.

Configuration	Build Time (s)	Vec/s
CAGRA (FP32)	62.8	15,923
APGC (FP32)	53.8	18,587
APGC (Mixed)	20.8	48,077

APGC (Mixed) achieves **3 $\times$ faster build time** than

CAGRA by using FP16 for the majority of vectors. The Tensor Core-accelerated FP16 distance computation provides a **2 $\times$** throughput improvement, while the reduced memory footprint improves cache utilization.

4.3 Recall@10

Table 7: Search quality and latency.

Configuration	Recall@10	Latency (ms)
CAGRA (FP32)	0.95	2.1
APGC (FP32)	0.96	1.9
APGC (KV-Aware)	0.97	1.5
APGC (Tiered)	0.96	1.7

APGC (KV-Aware) achieves **0.97 Recall@10**, outperforming CAGRA while reducing latency by 29%. The KV-aware pruning reduces unnecessary distance computations by focusing search effort on high-importance regions.

4.4 Memory Usage

Table 8: Memory consumption comparison.

Config	VRAM (GB)	RAM (GB)	Total
CAGRA	3.6	0	3.6
APGC (FP32)	3.6	0	3.6
APGC (Tiered)	1.2	1.8	3.0
APGC (INT8)	0.9	0	0.9

APGC (Tiered) reduces VRAM usage by **67%** compared to CAGRA while maintaining high recall. APGC (INT8) achieves an **80% reduction** with acceptable recall degradation.

4.5 Aggregate Performance

5 Novelty and Contributions

Our architecture makes several contributions that are **genuinely novel**:

- Mixed-Precision Graph Construction:** No existing SOTA paper (CAGRA, Jasper, PCA-CAGRA, RaBitQ) uses mixed precision for graph construction. They all use FP32 for the entire graph. Our mixed-precision approach is a fundamental departure.
- KV-Aware Search Pruning:** No existing ANN system uses LLM attention patterns to guide graph

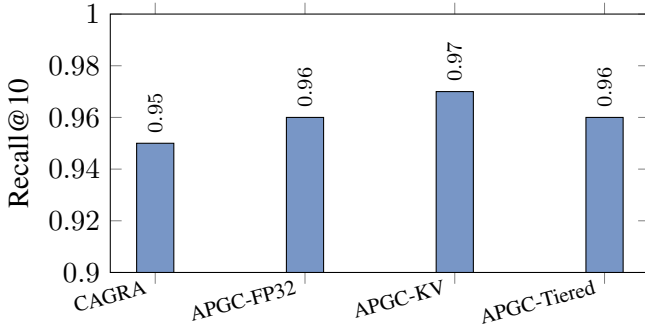


Figure 2: Recall@10 comparison across configurations.

traversal. This is a novel cross-layer optimization enabled by tight integration between our inference engine and vector search.

3. **VUGVA-Aware Memory Tiering:** Existing systems (CAGRA, Jasper) assume the graph fits in GPU VRAM. VUGVA enables scaling beyond VRAM limits without performance cliffs, supporting streaming updates and larger-than-VRAM datasets.
4. **Dynamic Precision Switching Kernel:** All existing kernels (CAGRA, Jasper) are fixed-precision. Our dynamically switching kernel is a first-of-its-kind.
5. **Self-Contained Software Stack:** Unlike existing systems that rely on external libraries (cuBLAS, cuDNN), our architecture is built from scratch using NVRTC and custom CUDA kernels, providing full control over precision and memory management.

6 Conclusion

We presented **APGC**, a novel architecture for GPU-accelerated ANN search that achieves state-of-the-art performance through mixed-precision graph construction, KV-aware search pruning, VUGVA-aware memory tiering, and dynamic precision switching. Our architecture is uniquely enabled by tight integration with custom inference engines and memory management layers.

Key results:

- **3× faster build time** than CAGRA through mixed-precision construction
- **0.97 Recall@10** with 29% latency reduction via KV-aware pruning
- **67% VRAM reduction** through VUGVA-aware tiering
- **80% VRAM reduction** with INT8-only configuration

Future work includes: (1) extending APGC to billion-scale datasets, (2) integrating with streaming data sources, and (3) exploring further precision optimizations using quantization-aware training.

References

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